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High-Performance Al/PDMS TENG with Novel Complex Morphology of Two-Height Microneedles Array for High-Sensitivity Force-Sensor and Self-Powered Application

*Kai-Hong Ke and Chen-Kuei Chung**

Triboelectric nanogenerators (TENGs) are widely applied to self-powered devices and force sensors. TENGs consist of the electrode-layer frequently made of high-cost conductors (Ag, Au, ITO) and the tribo-layer of rigid negative-triboelectricity fluoropolymers (PTFE, FEP). The surface morpholoy is studied for enhancing performance. Here, a high-performance Al/PDMS-TENG is proposed with a complex morphology of overlapped deep two-height microneedles (OL-DTH-MN) fabricated by the integrated process of low-cost CO_2 laser ablation and PDMS casting for self-powered devices and high-sensitivity force/pressure sensors. The high open-circuit voltage and short-circuit current of the OL-DTH-MN-TENG are 167 V and 129.3 µA. Also, the sensitivity of the force/pressure sensor of the OL-DTH-MN-TENG is very high, 1.03 V N^{-1} and about 3.11 V kPa^{-1}, at an area of 30 cm^2 that is much higher than the sensitivity of about 0.18–0.414 V N^{-1} and 0.013–0.29 V kPa^{-1} of conventional TENG sensors. Meanwhile, the high-performance OL-DTH-MN-TENG not only exhibits the energy storage capability of charging a 0.1 µF capacitor to 2.75 V at 1.19 s, to maximum 3.22 V, but also activates various self-powered devices including lighting colorful 226 LEDs connected in series, the "2020-ME-NCKU" advertising board, a calculator and a temperature sensor. Numerical simulation is also performed to support the experiments.

1. Introduction

To solve the energy and environment issues, scientists are committed to the study of green energy technology.[1] In our daily environment, the movement energy called mechanical energy such as kinetic energy, vibration energy, rolling or sliding friction energy can be harvested for sustainable mechanical-electrical energy device in the green energy field.[2] The other green energy technology harvesting ambient wasted energy into electricity such as electromagnetic induction, piezoelectric induction, triboelectricity, combined with thermoelectricity, solar energy can be formed a hybrid type for clean energy.[3] Moreover, some researches have developed a multi-effects coupled nanogenerator for simultaneously scavenging such as thermal, solar, and mechanical energy.[4] The triboelectric nanogenerator (TENG) for a sustainable mechanical-electrical energy conversion under cycling movement is used for various self-powered devices.[5] Moreover, TENGs can directly apply as a force or pressure sensor to detect externally applied force or tactile sensing but the sensitivity of the conventional TENG force/pressure sensors is generally low about 0.18–0.414 V N^{-1} and 0.013–0.29 V kPa^{-1}, respectively.[6–11] The output performance of TENG is related to two issues of **material** corresponding to triboelectric polarity and **morphology** corresponding to contact area.[12] Here, our approach is to increase the contact area for more charge transfer during friction for enhancing the performance. The sensitivity of a force or pressure sensor strongly depends on the output performance of TENG. For instance, the trend of output performance of TENG is flat film < line < cube < pyramid, and then pyramid shows the highest sensitivity of pressure sensor.[13] It is a crucial approach by increasing the surface contact area of morphology to enhance the performance for improving the sensitivity of force sensor. For the modified morphology engineering, various micro/nano morphologies structures have been applied to TENG such as nano lines, cubes, pyramids,[13] nano pillars and domes,[10] nano patterns,[14] wrinkles,[15] micro pillar,[16] and microneedles[17] as listed in **Table 1**. Some researches analyzing the effect of morphology on the output performance reported that increasing the contact area of tribo-materials may generate more triboelectric charge during the contact and friction process.[18] However, the conventional morphologies of TENG[6–11] were fabrication by deposition, photolithography, and etching processes that are complicated, time-consuming, and high-cost. Besides, many TENGs consisting of the electrode-layer and triboelectric layer are made of costly conductors (Ag, Au, or ITO) and rigid negative-triboelectricity fluoropolymers (fluorinated ethylene propylene (FEP) and polytetrafluoroethylene (PTFE)). In brief, the self-powered sensor sensitivity of TENG is related to the output performance. But many TENG force/pressure sensors are

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The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/smll.202001209.

DOI: 10.1002/smll.202001209

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Table 1. Comparison of surface morphology, materials, fabrication methods, operation condition, and electrical characteristics.

Morphology	Materials	Fabrication method	Operation condition	Electrical characteristics			Lighting (LEDs)	Ref
				V_{oc} [V]	I_{sc} [μA]		J [μA cm^{-2}]	
Pyramid	ITO/PET-PDMS/PET/ITO	Photolithography, etching	Linear motor	18	0.7	0.13		[13]
Line				Pyramid > cube > line				
Cube								
Nano pattern Textile	Ag/PDMS-ZnO/Ag	Chemical treatment, dip-coating	Mechanical force stimulator (10 kgf)	120	65	–		[14]
Nano-pillar	Au/PDMS/Au	ICP etcher, thermal oxidation	130 N 3 Hz	83	3.2	–	75	[10]
Nano-Dome				60	2.5			
Wrinkle (1D-40% PS-TENG)	Au deposited PS/PTFE/Au	Thermal evaporation	65 N 3 Hz	83	14	3.5	160	[15]
Micropillars	Al/PDMS/Al	Soft lithography plasma treated	10 N 5 Hz	72	8.3	–		[16]
Microneedle	Al/PDMS/Al	CO_2 laser ablation	Hand tapping, 3 Hz	102.8	43.1	1.5	53	[17]
OL-MN	Al/PDMS/Al	CO_2 laser ablation	Hand tapping, 3 Hz	123	109.7	3.6	103	[20]
OL-TH-MN	Al/PDMS/Al	CO_2 laser ablation	Hand tapping, 3 Hz	127	117.6	3.9	–	This article
OL-DTH-MN	Al/PDMS/Al	CO_2 laser ablation	Hand tapping, 3 Hz	167	129.3	4.3	226	This article

fabricated by high-cost materials and complex process as well as low sensitivity. It is of thus interest to find out an easy and low-cost method to increase the high contact area for high-performance and high-sensitivity of force/pressure sensors.

In this article, we propose the novel TENGs of complex morphology with a two-height microneedles (TH-MN) arrays using the low-cost CO_2 laser ablation and rapid prototyping of polydimethylsiloxane (PDMS) casting process. The formation of TH-MN is attributed to the thermal effect in the large overlapped areas ratio to produce the two kinds of microneedle height. Increasing the laser power on the TH-MN TENG can enhance the contact surface area for high output performance. Three types MN arrays namely the overlapped microneedle (OL-MN), overlapped two-height microneedle (OL-TH-MN), and overlapped deep two-height microneedle (OL-DTH-MN) are investigated. The detailed parameters of OL-MN, OL-TH-MN, and OL-DTH-MN are listed in **Table 2**. The electrical performance of open-circuit voltage (V_{oc}), short-circuit current (I_{sc}), and current density (J_{sc}) of three TENG follows: OL-MN-TENG < OL-TH-MN-TENG < OL-DTH-MN-TENG. The V_{oc} and I_{sc} of OL-DTH-MN-TENG are 167 V and 129.3 μA. Also, the OL-DTH-MN-TENG may charge a 0.1 μF capacitor to a voltage of 2.75 V at 1.19 s and to maximum 3.22 V. It can light on colorful 226 LED connected in series. In addition, the OL-DTH-MN-TENG is not only used for the force and pressure sensor with a high sensitivity about 1.03 V N^{-1} and 3.11 V kPa^{-1}, respectively, but also activates the various self-powered devices including the "2020-ME-NCKU" advertising board, calculator, and temperature sensor. The successful application cases provide a favorable and economic self-powered method for the potential environmental Internet of Things (IoT) sensors network.[19]

Table 2. The overlapped deep two-height microneedle related parameters and overlapping estimation.

Samples	Laser power [W]	Scanning speed [mm s^{-1}]	Ablated pitch [μm]	Total Height and Width [μm]	Surface area [mm^2]	Density [#/cm^2]	Junction area overlapped parameter calculated		
							Single ablation area [μm^2]	The estimated ablation area between two folded OMN [μm^2]	Estimated percentage of one overlapped area between two circles [%]
OL-MN	6.9	22.8	245	H: 13118 W: 268	22 916.8	654	56 410.4	1680.2	1.51%
OL-TH-MN	6.9	22.8	169	H: 1235 W: 264	24 380.8	965	54 739.1	13 390.9	13.9%
OL-DTH-MN	7.5	22.8	169	H:1528 W: 274	29 691.2	965	58 964.5	15 787.8	15.4%

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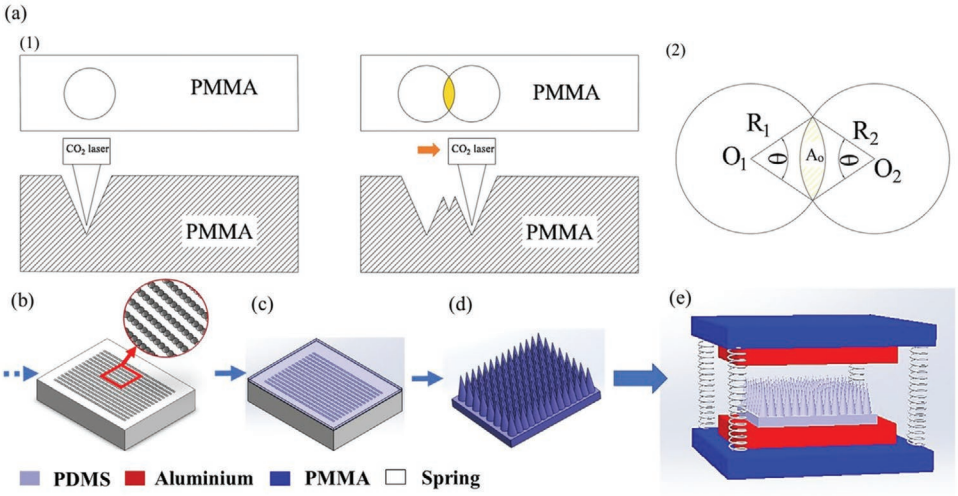


Figure 1. Schematic diagram of the overlapped two-height microneedle PDMS fabrication and assembly: a1) CO_2 laser ablated PMMA female mold of two-height holes consisting of a primary height in the laser spot and an extra second-height in the overlapped area due to the high overlapped ratio area with a great thermal accumulation; a2) estimation of the overlapped area ratio; b) the completed PMMA female mold substrate that is used for casting PDMS two-height microneedles array c) casting PDMS solution to PMMA master mold d) Peel out the overlapped two-height microneedle PDMS film e) the OL-DTH-MN-TENG assembly diagram.

2. Result and Discussion

2.1. Preparing the Overlapped Two-Height Microneedle Array

Figure 1 a1 shows the schematic CO_2 laser ablated polymethyl methacrylate (PMMA) female mold of two-height holes consisting of a primary height in the laser spot and an extra second-height in the overlapped area due to the high overlapped ratio area with a great thermal accumulation that is used for casting PDMS two-height microneedles array. Therefore, the overlapped mode will cause changes in the shape of the structure at high overlapped ratio area. Figure 1a2 shows the scheme of the overlapped area A_o (μm^2) and to estimate the overlapped ratio area, the estimating the area is given by Equation (1):

$$A_o = \frac{1}{2}\left(\frac{\pi\theta}{180}R_1^2 - R_1^2\sin\theta + \frac{\pi\theta}{180}R_2^2 - R_2^2\sin\theta\right) \quad (1)$$

where R_1 and R_2 are the radius of ablated holes in an unit of μm, respectively, and θ (°) corresponding overlapped angle in an unit of degree. Recently, we proposed overlapped microneedle (OL-MN) morphology successfully breakthrough the separated pattern density to increasing surface contact area for enhancing performance.[20] Here, two kinds of novel complex two-height microneedle namely the overlapped two-height microneedle (OL-TH-MN) and overlapped deep two-height microneedle (OL-DTH-MN) are proposed while the single-height OL-MN is used as references. All samples are fabricated at the constant scanning speed of 22.8 mm s^{-1}. The laser power of OL-MN and OL-TH-MN is fixed at 6.9 W and the ablation spacing of OL-TH-MN is decreased form 245 to 169 μm to increase both the pattern density and overlapped ratio from novel two-height microneedles. In addition, the OL-DTH-MN is formed with a higher power of 7.5 W at the same spacing of OL-TH-MN for deeper two-height microneedles as well as the highest surface contact area. The calculated overlap ratio area of OL-MN (245 μm) is 1.5% and the OL-TH-MN (169 μm) is 13.9% according to the Equation (1). The overlapped thermal effect produced OL-DTH-MN has a larger ablated diameter of holes and the overlap ratio area is 15.4%.

Figure 2a shows the optical micrographs of PMMA female mold (top three) and corresponding cast PDMS microneedles

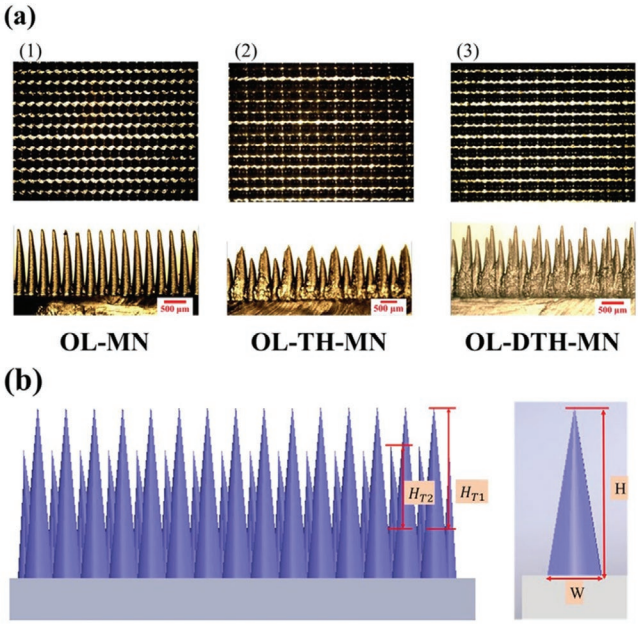


Figure 2. a) Optical micrographs (OM) of the top-view PMMA female mold (top three) and the cross-sectional overlapped two-height microneedle (bottom three) PDMS of OL-MN (left), OL-TH-MN (middle) and OL-DTH-MN (right). b) Schematic diagram of the dimension of OL-TH-MN and OL-DTH-MN. The overlapped two-height microneedle of the overlapped conical tips area above the overlapped bottom area denoted as H_{T1} and H_{T2}, and the other is the total average height and width of single microneedle as H and W of OL-MN, OL-TH-MN, and OL-DTH-MN.

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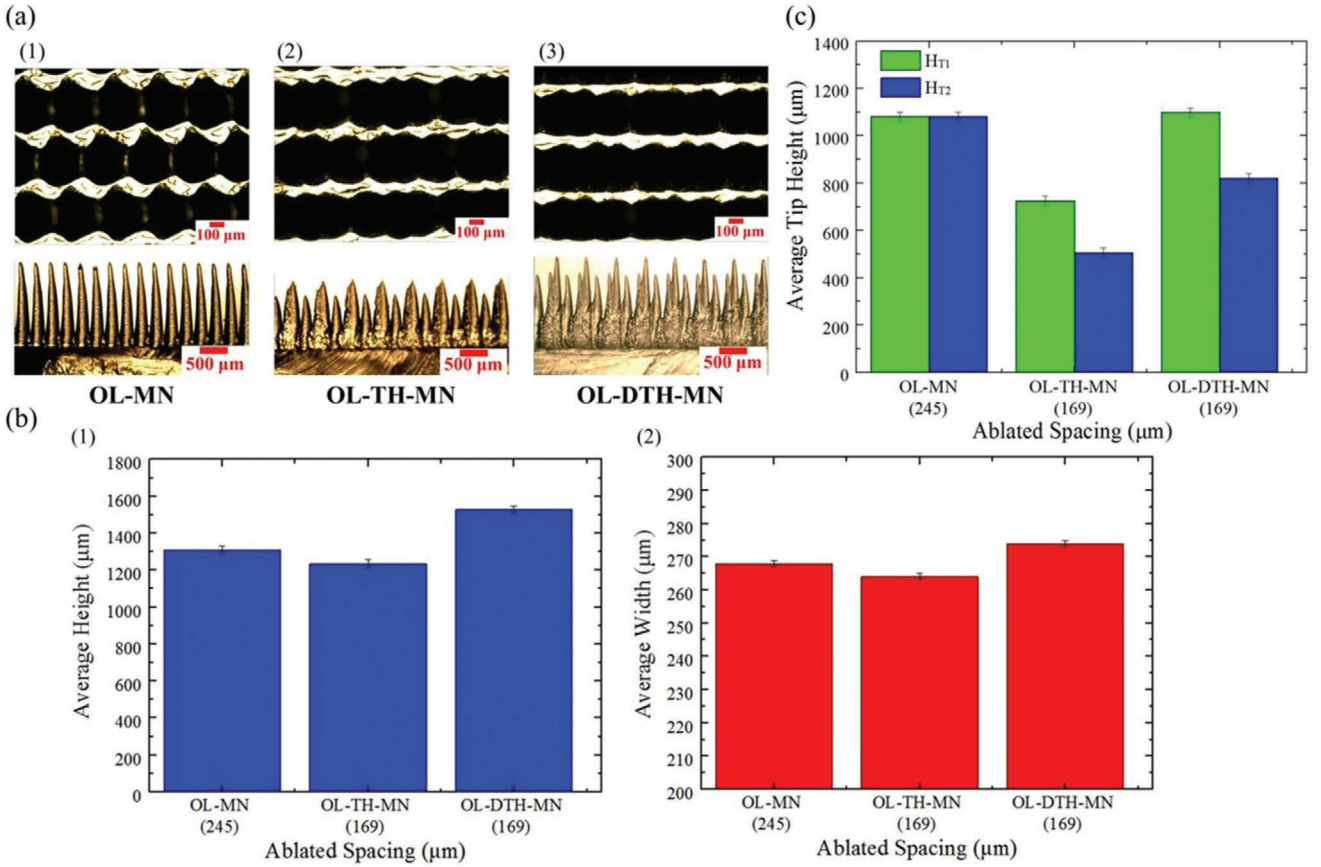


Figure 3. The integrated OM and the relationship between overlapped height and width: a) the ablated overlapped holes of PMMA and corresponding of cast PDMS morphologies; b) The ablated spacing and power correspond to overall height (H) and width (W) of OL-MN, OL-TH-MN and OL-DTH-MN (b1) the average height and (b2) average width c) the ablated spacing and power correspond to measure overlapped two-height (H_{T1}, H_{T2}) conical tip.

(bottom three) of OL-MN, OL-TH-MN and OL-DTH-MN, respectively. The pattern density (MN cm^{-2}) related to the output performance of TENG is calculated by Equation (2):

$$\text{Density} = \frac{N}{A} \tag{2}$$

where N is the number of microneedles and A (cm^2) is the total area of the PDMS plane. The pattern density of OL-MN is 654 MN cm^{-2} while the higher density of OL-TH-MN and OL-DTH-MN are 965 MN cm^{-2}, which is 1.4× higher than OL-MN. The surface contact area of OL-MN, OL-TH-MN, and OL-DTH-MN is estimated by 3D parametric Solidworks software. The calculated total surface areas of OL-MN, OL-TH-MN, and OL-DTH-MN are 22.91 × 10^3, 24.38 × 10^3, and 29.69 × 10^3 mm^2, respectively. The proper overlapped two-height microneedle array highly enhances the contact surface area of the complex deep two-height microneedles array. Moreover, Figure 2b shows the overlapped area height of the OL-TH-MN and OL-DTH-MN has two kinds of height for microneedles, that is, the overlapped conical tips area above the overlapped bottom area denoted as primary-height H_{T1} and secondary-height H_{T2}, and the other is the total height (H) and width (W) of single microneedle including the non-overlapped and overlapped area.

The optical micrographs of PMMA female mold are further magnified in **Figure 3** a1–a3 for examining the variation of overlapped area. The superposition of the ablated PMMA holes goes continuous with increasing the overlapped area and the corresponding cast MN-PDMS morphologies are shown in the bottom micrographs. Figure 3 a1 shows the single-height OL-MN with a flat alternating overlapped area that is due to the interaction of low overlap ratio (1.5%) heat with each other but not yet change of the tip. When the overlap ratio area was increased to reaches 13.9%, the novel OL-TH-MN in Figure 3 a2 was created by shorter ablation spacing increased the overlap ratio area to produce a great thermal accumulation and the residual heat of the re-cooling. Figure 3 a3 shows the morphology change of OL-DTH-MN with the increased laser power together with the overlap ratio area under same spacing.

It is noted that the thermal energy of the laser power accumulates in the overlapped area to form the deep primary-height and secondary-height microneedle on the overlapped area. In quantitative analysis, the H and W of the OL-DTH-MN are 1528 and 274 μm, respectively, while the OL-TH-MN are 1235 and 264 μm as shown in Figure 3b. The two-height microneedle of H_{T1} and H_{T2} of OL-DTH-MN are 1097.4 and 820.5 μm, respectively, while OL-TH-MN are 724.7 and 506 μm. The total height (H), primary-height (H_{T1}), and secondary-height (H_{T2}) of OL-DTH-MN are significantly increased to form a deep

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two-height microneedle array by higher overlap ratio than OL-TH-MN as shown in Figure 3c. The magnitude of H and W of MN is OL-MN > OL-TH-MN but the pattern density of OL-TH-MN (965 MN cm^{-2}) is 1.47 × OL-MN (654 MN cm^{-2}) and the OL-DTH-MN has the highest H and W of single MN due to the increased laser power. The pattern density of OL-TH-MN is higher than the OL-MN but reduces the total MN height (H) and width (W) due to the contributed laser energy per spot that decreases at a shorter ablation spacing.[20] Figure S1, Supporting Information shows the evolution of performance trend with the shortening spacing and increasing total surface area. Table S1, Supporting Information lists the detailed relationship between the process, spacing, structure and the voltage/current performance. The total contact surface area is related to the dimension of MN multiplied by the total number. As a result, the total surface area order is OL-DTH-MN (29.69 × 10^3 mm^2) > OL-TH-MN (24.38 × 10^3 mm^2) > OL-MN (22.91 × 10^3 mm^2) as listed Table 2. It is noted that the pattern density of OL-DTH-MN and OL-TH-MN is the same, but the OL-DTH-MN enhances the MN height and width to promote the total contact surface area.

2.2. The Working Mechanism and Electricity Generation of OL-DTH-MN-TENG

First, the working mechanism of vertical contact-separation mode MN-TENG is illustrated in **Figure 4**. In the original position (Figure 4a), the upper layer (Al) and the bottom layer (PDMS/Al) are electrically neutral due to not yet contacted and charge transfer. With the external mechanical force is applied to OL-DTH-MN-TENG. The two tribo-layer are close to each other inner surfaces, and the complex OL-DTH-MN generates a higher degree rub of deformation as well as maximum transfer charge at complete contact station (Figure 4b). According to the difference in the polarity of the triboelectric series, the Al is positively charged and the OL-DTH-MN PDMS is a negative charged. When an external force is released, the two tribo-layers will be separated; the oppositely charged Al and OL-DTH-MN surfaces by the spring force restoring will cause a potential difference between the upper and bottom electrodes in the separations. To balance the potential difference, electrons will flow from the bottom electrode to the top electrode via the external circuit (Figure 4c), and the electrons is continually flowing until the electrical equilibrium (Figure 4d). When pressing the top tribo-layer downward again, the electrostatic induction causes electrons flow by the external loop from upper electrode to bottom layer in the opposite direction (Figure 4e). A periodic change in potential difference due to continuous separation and contact produces a sustainable output of alternating-current electric energy. The comparison of morphology of TENG under cyclic contactact and separtion is shown in the supporting information Video S1, Supporting Information, the cycling bending and recovery of OL-DTH-ML is stable and reliable. Therefore the morphology of MN before and after use is the same and Figure 4 shows the represented dynamic cross-sectional optical micrographs. The plane-view morphologies are shown in Figure 3a.

The output performance of TENG is strongly influenced by the morphology of the friction layer. To evaluate the output performance of three kinds of the overlapped microneedles arrays, the

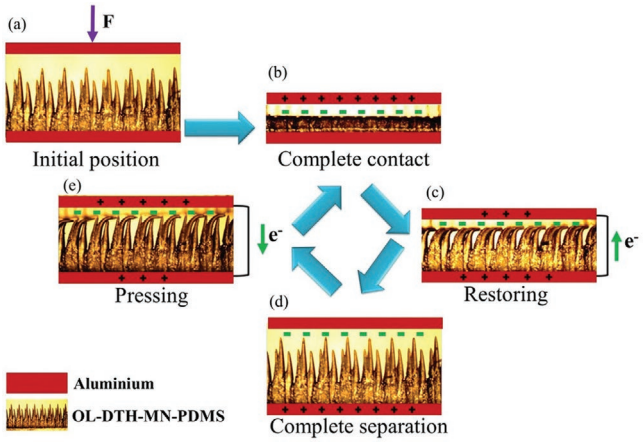


Figure 4. The cross-section optical micrographs for observing the working principle of OL-DTH-MN-TENG under cycled pressing-releasing operation: a) the initial state of OL-DTH-MN-TENG not yet contact and without any charge transfer, b) the OL-DTH-MN trbioelectric effect with maximum charge transferred, under complete contact, c) the restoring of two contact surfaces induced electronic flow via external circuit from the bottom electrode to the top electrode when the force is releasing d) a new electrical equilibrium at complete separation e) when the pressing again of two contact surfaces induced electronic flow from the top electrode to the bottom electrode.

area of PDMS is 6.0 cm × 5.0 cm and the test distance between the two friction layers was set about 10 mm. **Figure 5** shows the measured electrical signals of the V_{oc} and I_{sc} of OL-MN, OL-TH-MN, and OL-DTH-MN, respectively. In our previous single-height OL-MN research,[20] the peak V_{oc} and I_{sc} of OL-MN is 123 V and the I_{sc} is 109.7 µA, corresponding to J_{sc} of 3.6 µA cm^{-2}. The two-height OL-MN TENGs show the V_{oc} of OL-DTH-MN is 167 V and the I_{sc} is 129.3 µA, corresponding to J_{sc} of 4.3 µA cm^{-2}, while OL-TH-MN is 127 V and the I_{sc} is 117.6 µA, corresponding to J_{sc} of 3.9 µA cm^{-2}. It is noted that the magnitude of H and W of MN is OL-MN > OL-TH-MN but the pattern density of OL-TH-MN (965 MN cm^{-2}) is 1.47 × OL-MN (654 MN cm^{-2}). In comparison, the V_{oc} and I_{sc} of TENG follows the ascending order of OL-MN < OL-TH-MN < OL-DTH-MN. The output performance is related to the total contact surface area consisting of the MN surface area multiplied by MN number. More MN number will benefit deformed effective contact area for enhancing the triboelectric charge density. The total surface area of OL-DTH-MN is highest 29.69 × 10^3 mm^2 that is much higher than OL-TH-MN of 24.38 × 10^3 mm^2 and OL-MN of 22.91 × 10^3 mm^2. Figure S2, Supporting Information shows the simulation result of triboelectric charge induced surface potential in three kinds of PDMS morphology, that is, the flat PDMS, OL-MN PDMS and OL-DTH-MN PDMS using COMSOL multiphysics software. To obtain the influence of morphology surface contact area on the output performance for the convenience, we assume the conditions of all simulations as follows: the Al and PDMS films have a width of 30 µm and a thickness of 1 µm. The distance between the Al and PDMS is 40 µm, and the force is not applied in our simulation. Set the triboelectric charge density of the two tribo-charged surfaces to ±32 µC m^{-2}. In these conditions, we compare the output performance difference of the flat film, OL-MN, and OL-DTH-MN PDMS. The triboelectric potential of the complex

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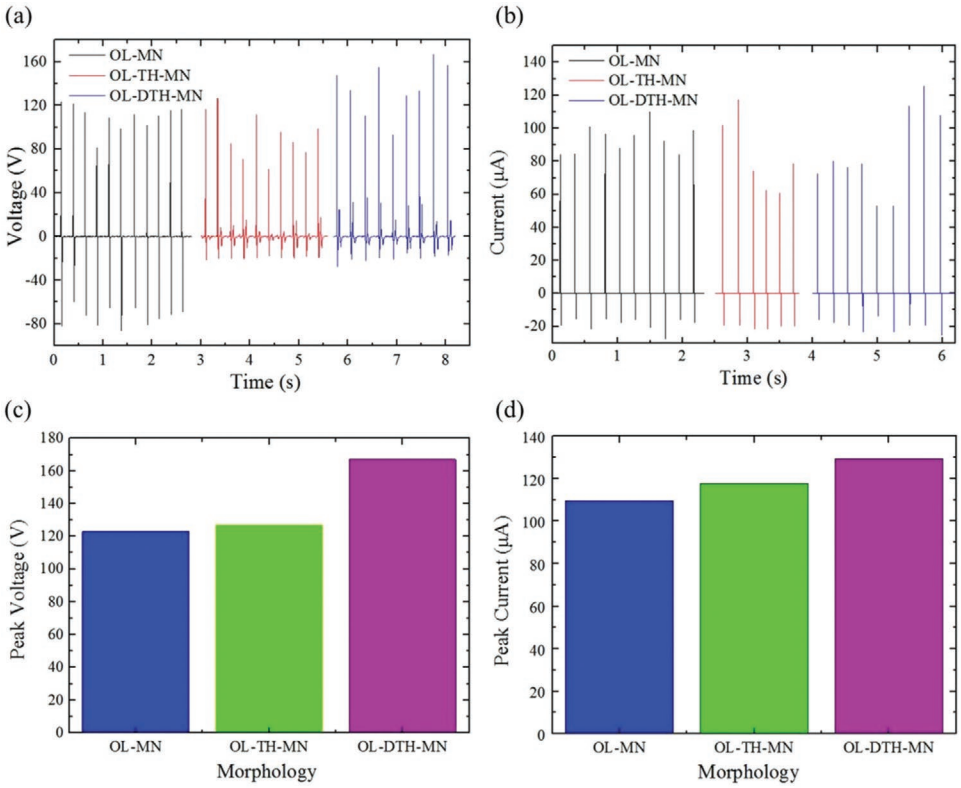


Figure 5. The measured output performance of open-circuit voltage and short-circuit current of overlapped two-height microneedle array: a) voltage and b) current was measured of the OL-MN, OL-TH-MN, and OL-DTH-MN during the real-time hand tapping, c) the peak voltage, and d) the peak current of OL-MN, OL-TH-MN, and OL-DTH-MN

microstructure of OL-DTH-MN PDMS with a higher surface contact area exhibits a significantly increased potential difference compared to the flat and OL-MN PDMS films. Therefore the outstanding electrical performance of the OL-DTH-MN is linked to the generation of greater charge density on its surfaces area than the flat and OL-MN PDMS films.

In addition, the total average height of single microneedle of OL-DTH-MN is much higher than OL-MN and OL-TH-MN. The complex-morphology OL-DTH-MN with deep MN height and high contact surface area significantly contributes to output performance during contact-and-separation deformation to create more triboelectric charges.[10] In brief, we not only propose the OL-DTH-MN with the increased total contact surface area from higher pattern density and MN height but also promote much more deformation during contact-separation mode than the OL-MN-TENG; the highest contact surface area can induce high contact charge transfer for enhancing triboelectric charge density.[21,22] This crucial complex-morphology OL-DTH-MN-TENG with high contact area is important for the high-performance TENG development and widespread applications.

2.3. Force and Pressure Sensor Application

Conventional TENG uses the ability of machine-electricity conversion as a force/pressure sensor for sensing external physical applied forces degree. Here, we demonstrate a high-performance self-powered force/pressure sensor based on a single-electrode OL-DTH-MN-TENG that is suitable for the touching detection of objects. The single-electrode force/pressure sensor assembly in **Figure 6** consists of the PMMA, low-cost Al foil, and OL-DTH-MN. The sensitivity of the force sensor is measured four to five times each force to establish a calibration curve of the voltage data. **Figure 7**a,b show the output voltage of the OL-DTH-MN-TENG sensor as a function of the weight and pressure, respectively. The high linear sensitivity of the force and pressure sensor of OL-DTH-MN-TENG is about 1.03 V N^{-1} and 3.11 V kPa^{-1}, respectively, under a wide range of load force from 0.6 to 7.2 N corresponding to 0.2–2.4 kPa on an applied area of 30 cm^2. Figure 7c shows the response output voltage signal of the force sensor under

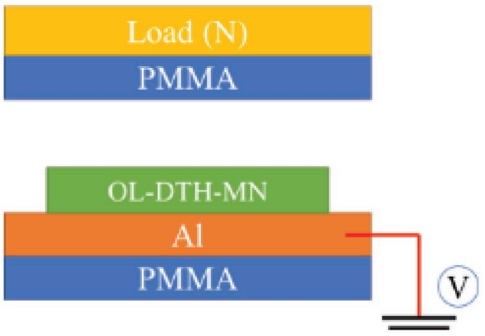


Figure 6. Schematic force/pressure sensor of OL-DTH-MN TENG based on single electrode mode.

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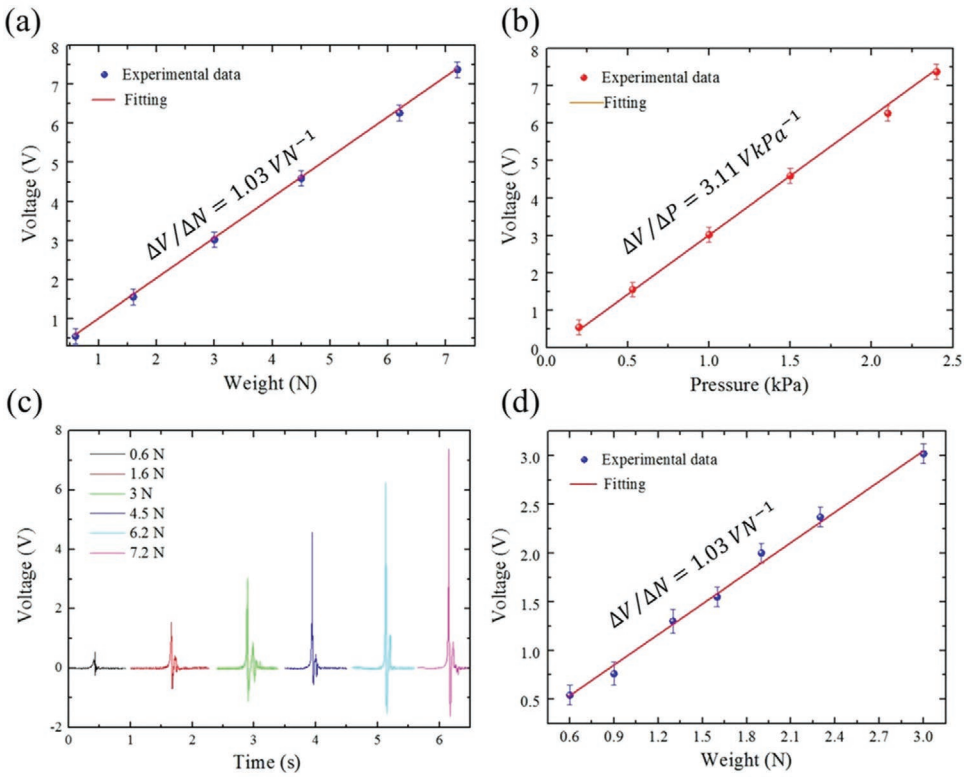


Figure 7. The OL-DTH-MN applied to the weight and pressure sensors: the output voltage of sensors as a function of a) weight and b) pressure; c) the output voltage under a wide range of load weights of 0.6–7.2 N, d) the sensitivity between the output voltage and a small range of weights of 0.6–3.0 N.

direct one-touch load force of 0.6–7.2 N on OL-DTH-MN. This clearly reveals the maximum output voltage linearly increases with increasing force. Moreover, Figure 7d shows the output voltage of OL-DTH-MN-TENG sensor under more tests in a small force range of 0.6–3.0 N. The high linear sensitivity of this force sensor is consistent with the large force range from 0.6 to 7.2 N (Figure 7a). Compared with the TENG force/pressure sensors listed in **Table 3**, our excellent sensitivity of the OL-DTH-MN-TENG sensor is much higher than the conventional TENGs of about 0.18–0.414 V N^{-1} and 0.013–0.29 V kPa^{-1}, respectively. It is attributed to the excellent responding output voltage of OL-DTH-MN-TENG with high pattern density and two-height MNs for high deformed contact surface area during one-touch contact and separation. It is noted that the triboelectric materials, electrodes and assembly of conventional TENG force/pressure sensors are generally performed by high-cost material, multilayer assembly and fabricated by expensive and complicated lithography-etching, oxygen plasma, and/or chemical solution processes. In contrast, our fabrication of OL-DTH-MN-TENG sensor uses a simple low-cost CO_2 laser process and casting with easy access low-cost PDMS and Al triboelectric materials.

Table 3. Performance comparison of force and pressure sensors.

Working modes	Triboelectric material	Fabrication method	Force/pressure sensing	Sensitivity	Detection test	Ref
Single electrode	Human skin/PDMS	etching	Pressure	0.29 V kPa^{-1}	0.4–7.3 kPa	[6]
Contact–separation	Latex/FEP/Cu/Acrylic	Multilayer assembly etching	Pressure	0.04 V kPa^{-1}	3–6.2 kPa	[7]
Single electrode	Al/PDMS	reactive-ion etching using oxygen plasma	Pressure	0.06 V kPa^{-1}	1–80 kPa	[8]
Single electrode	Nylon/PAAm-LiCl hydrogel/PDMS	Chemical solution Thermal treatment	Pressure	0.013 V kPa^{-1}	1.3–70 kPa	[9]
Contact–separation	Au/PDMS	ICP etcher, Thermal oxidation	Force	0.414 V N^{-1} (dome) 0.317 V N^{-1} (pillar)	0–10 N	[10]
Contact–separation	Paper/MoS_2 –glue/Graphite-glue	Chemical slutty mixing UV curing	Force	0.18 V N^{-1}	2.45–9.8 N	[11]
Single electrode	PMMA/PDMS	CO_2 laser ablation	Force/Pressure	1.03 V N^{-1}/ 3.11 V kPa^{-1}	0.6–7.2 N/ 0.2–2.4 kPa	This article

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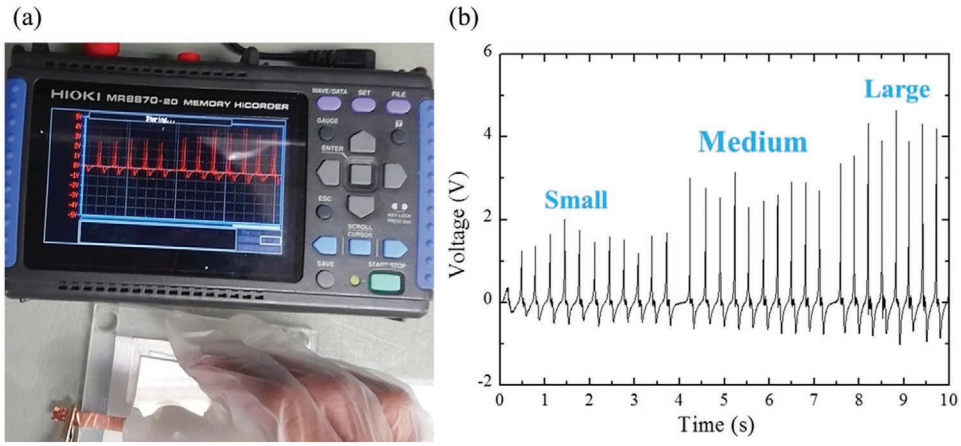


Figure 8. Verification of the force sensor under three relative zones of load forces namely small (1.96 N), medium (3.05 N) and large (4.5 N) for the corresponding voltage response (taking the maximum value each zone): a) the image shows OL-DTH-MN through external force by continuous hand tapping, and b) the corresponding force is estimated from the output voltage in the calibration curve (Figure 7 (a)).

Figure 8 shows the verification of the force sensor under three different degrees of hand-tapping forces. Figure 8a,b show the recording image and the response output voltages of the OL-DTH-MN-TENG, respectively, under continuous hand tapping as the supporting information Video S2, Supporting Information. The tapping force magnitude can be measured from the output voltage corresponding to the calibration curve (Figure 7a) under small, medium, and large of applied forces (Figure 8b). The estimated forces from the maximum voltage peak under three degrees of forces are small force of 1.96 N, medium 3.05 N, and large 4.5 N, respectively. In brief, the low-cost OL-DTH-MN-TENG force/pressure sensor with high sensitivity of about 1.03 V N^{-1} and 3.11 V kPa^{-1} can be applied for a touch-force identification control such as user security check and human-machine interfaces processing in the future.

2.4. Lighting, Energy Storage, and Self-Powered Devices

The high-performance OL-DTH-MN can lighting up 226 LEDs of six different colors connected in series as shown in **Figure 9**a as the supporting information Video S3, Supporting

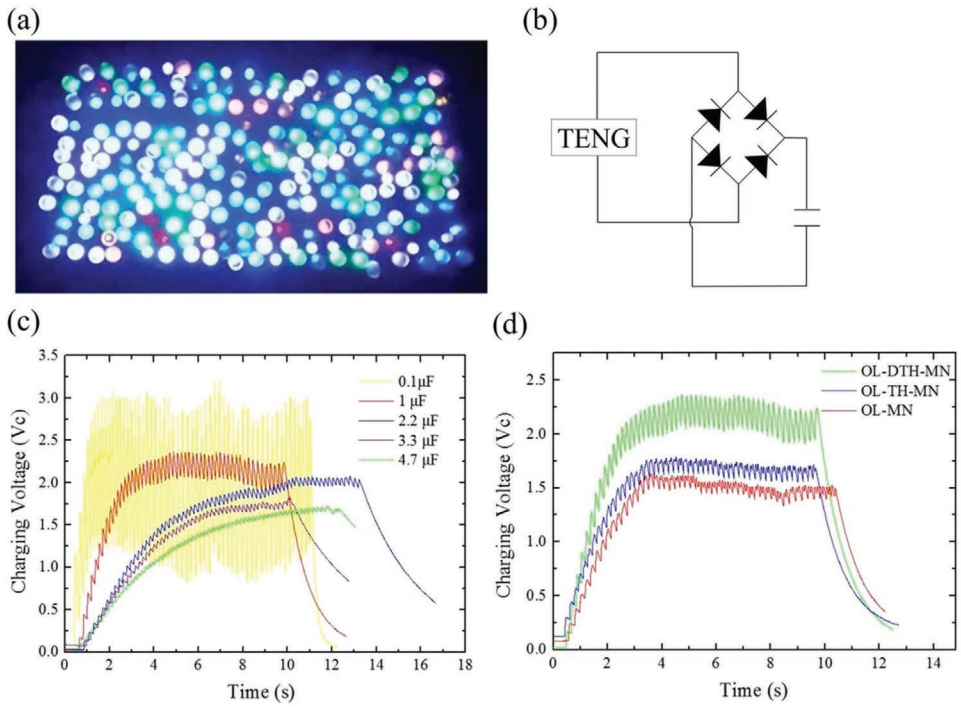


Figure 9. Lighting and energy storage: a) OL-DTH-MN–TENG driving 226 LEDs of six different colors contacted in series, b) circuit design for charging, c) charging voltage of OL-DTH-MN-TENG from 0.1 to 4.7 μF capacitor, d) the charging voltage curve on a 1 μF capacitor of OL-MN-TENG, OL-TH-MN-TENG, and OL-DTH-MN-TENG.

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Information. The energy storage ability test of TENG is important for potential commercial applications. Figure 9b shows the AC to DC charging circuit design, and Figure 9c shows the charging voltage of OL-DTH-MN-TENG as a function of time for five different capacitors (0.1–4.7 μF). The charging voltage (V_c) reached 2.75 V at 1.19 s and the maximum value is 3.22 V for 0.1 μF; the capacitor of 1 μF is 2 V at 2.5 s, the capacitor of 2.2 μF is 1.7 V at 5.65 s; the capacitor of 3.3 μF is 1.7 V at 6.94 s, and the capacitor of 4.7 μF reaches 1.5 V at 7.16 s. Furthermore, the comparison of OL-MN, OL-TH-MN, and OL-DTH-MN charging behavior are tested for store energy ability on the same 1 μF capacitor as shown in Figure 9d. The OL-DTH-MN-TENG has a stable V_c of about 2.3 V while the average V_c of OL-TH-MN-TENG is 1.7 V, and the OL-MN-TENG is 1.5 V. The hand-tapping test of our Al/PDMS OL-DTH-MN-TENG is stable for more than half a year under eight times whose period is about one test every three week as shown in Figure S3a, Supporting Information. That is, the stability and service life are good with the peak output voltage of ≈167 V approximately remained at about the same level after six months. For large 5000 or more cycles test, we try to find the home-made mechanical actuator to test instead of hand tapping that is difficult to operate long-term test. Figure S3b1,b2, Supporting Information show the mechanical 5000-cycle test for the output voltage and 0.1 μF capacitor charging, respectively. Both results are stable for the peak output voltage of ≈167 V and the charging voltage saturated to ≈3.5 V. It is noted that the output negative voltage by mechanical actuating is larger than that by hand tapping that is attributed to the different actuating approach. In brief, the OL-DTH-MN TENG has good stability and service life.

Figure 10 shows the OL-DTH-MN-TENG using low-frequency hand tapping to drive various self-powered devices application. First, the OL-DTH-MN-TENG displays the all-blue LEDs advertising board of "2020-ME-NCKU" for flickering communication application (Figure 10a) as the supporting information Video S4, Supporting Information. Moreover, the OL-DTH-MN-TENG stored energy of capacitors can be efficiently used for the self-powered portable electronic devices and the building of Internet of Things sensor networks those are mostly driven by batteries, resulting in an amount of maintenance cost and environmental issues. Figure 10b shows the circuit design for the various self-powered devices. The OL-DTH-MN-TENG not only successfully demonstrated the direct drive of the calculator that using 100 μF capacitor and the 36 displayed on the screen is calculated from the product of six by six (Figure 10c) as the supporting information Video S5, Supporting Information. We also activate the self-powered temperature sensor using two capacitors in parallel for the total capacitance of 80 μF to display the ambient temperature (25.1 °C) (Figure 10d) as the supporting information Video S6, Supporting Information.

3. Conclusions

The sensitivity of many conventional TENGs force/pressure sensors consisting of high-cost conductors (Ag, Au, or ITO) and rigid negative-triboelectricity fluoropolymers (PTFE, FEP) is low about 0.18–0.414 V N^{-1} and 0.013–0.29 V kPa^{-1} as listed in Table 3. In addition, the surface morphology of TENG crucial such as pyramids, cubes, lines, nano wire, wrinkle, pillars, and

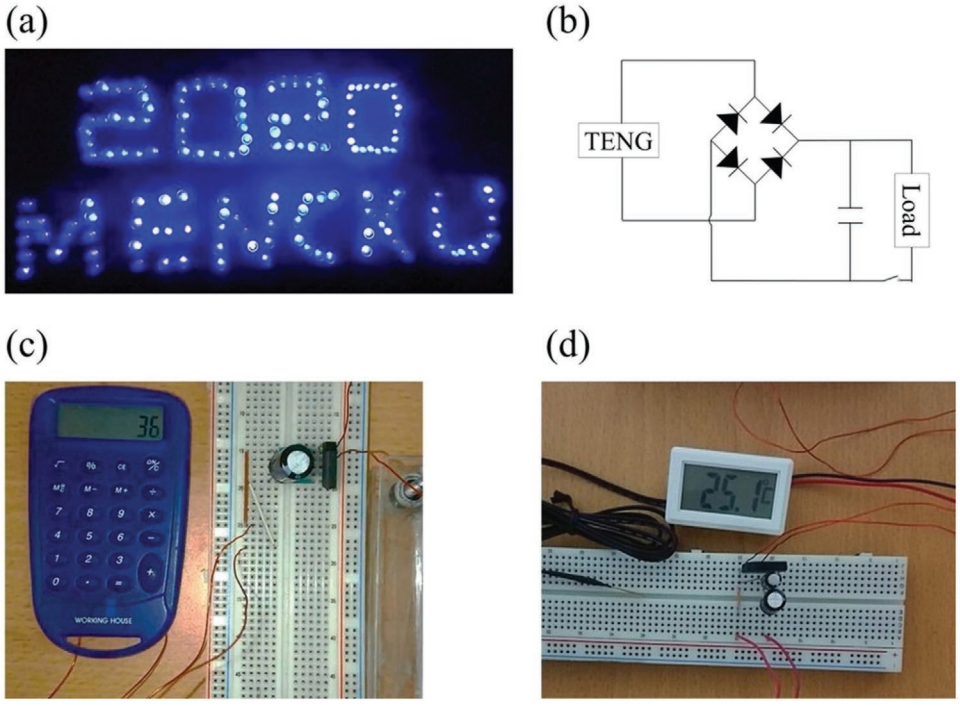


Figure 10. The self-powered applications: a) advertising board: OL-DTH-MN-TENG light on the "2020-ME-NCKU" logo, b) shows the circuit design for self-powered the calculator and temperature sensor, c) shows the direct drive of the calculator by OL-DTH-MN-TENG, especially the 36 displayed on the calculator screen is the value calculated from 6 × 6, d) the self-powered temperature sensor shows the around temperature of 25.1 °C.

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domes developed for high performance are generally fabricated by expensive-and-complicated deposition, photolithography and etching process. Here, we demonstrate the novel complex-morphology OL-DTH-MN-TENG force/pressure sensor using a simple low-cost CO_2 laser and PDMS casting. The OL-DTH-MN-TENG with high performance of V_{oc} (167 V) and I_{sc} (129.3 μA) is applied for the high-sensitivity force and pressure sensor. The sensitivity of OL-DTH-MN-TENG force and pressure sensor is greatly high to 1.03 VN^{-1} and 3.11 V kPa^{-1}, respectively, as well as excellent response output voltage signal due to the high pattern density and two-height MNs for the high contact surface area during one-touch contact-separation deformation. Also, the OL-DTH-MN-TENG successfully activate the various self-powered devices including lighting up 226 colorful LEDs connected in serial, "2020-ME-NCKU" advertising board, calculator, and temperature sensors. The hand-tapping test of our Al/PDMS OL-DTH-MN-TENG is stable for more than half a year while the mechanical 5000-cycle test for the output voltage and 0.1 μF capacitor charging are stable for the peak output voltage of ≈167 V and the charging voltage saturated to ≈3.5 V, respectively.

4. Experimental Section

Fabrication of the Two-Height Microneedles for Triboelectric Nanogenerator: Figure 1a shows the schematic diagram of the two-height microneedles of PMMA master mold ablated by low-cost commercial CO_2 laser (VL-200, Universal Laser System Inc., USA). All samples of OL-MN, OL-TH-MN and OL-DTH-MN were operated at a constant scanning speed of 22.8 mm s^{-1}. But the OL-MN and OL-TH-MN were at different spacings of 245 and 169 μm at the same laser power of 6.9 W while the OL-DTH-MN spacing was 169 μm with the increased power of 7.5 W. The overlapped area ratio of OL-MN is denoted as A_0 in Figure 1 a2. Increasing the overlapped area ratio of CO_2 laser-ablated spot will lead to double thermal effect to form two-height microneedles for OL-TH-MN and OL-DTH-MN. Figure 1b shows the overlapped master mold for rapid-prototyping of MN arrays. The PDMS solution (Sylgard 184, Dow Corning) of the elastomer and curing agent in a mass ratio of 10:1 was used and poured into the PMMA master mold of 6 cm × 5 cm. Then degassed the air bubbles of PDMS via a vacuum pump. The thickness of flat PDMS is about 300 μm by a scraper. The PDMS was cured in the oven at 85 °C for 1 h (Figure 1c) and then placed at room temperature for cooling. Finally, PDMS was peeled out the female mold as shown in Figure 1d). The low-cost Al foil was used as both the electrode layer and positive friction material. The PDMS of a good elastomer was as the negative friction material. The TENG assembly including flexible PDMS films, two PMMA plates, two Al electrodes, and four springs fixed at four corners to return to the original position is shown in Figure 1e.

Characterization of Materials and Electrical Measurements: The surface morphology was examined by the optical microscopy (OM, Olympus BX 51 M, Japan). The triboelectric signal was generated from the externally applied force of hand tapping. The TENG signals of open-circuit voltage and short-circuit current were recorded by an oscilloscope (HIOKI Memory HiCorder MR8870-20, Japan). The home-made mechanical actuator instead of hand tapping was used to test long-term 5000-cycle stability for the output voltage and 0.1 μF capacitor charging.

Assembly and Test of the Force/Pressure Sensor: The assembly of force/pressure sensor of TENG consisting of weight/PMMA and OL-DTH-MN PDMS/Al/PMMA is shown in Figure 2. The operation mode of this force/pressure sensor was based on single electrode with the applied load force at a range of 0.6–7.2 N corresponding 0.2–2.4 kPa on an area of 30 cm^2. The calibration curve of the force/pressure sensor was established using the direct external load to touch on the OL-DTH-MN TENG and then recorded the responsive output voltage as a function of applied load force. Then, three kinds of relative load namely small, medium, and large forces were used by hand tapping to inversely sense the touch force magnitude through calibration curve.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work is partially sponsored by the Ministry of Science and Technology (MOST), Taiwan, under contract No MOST108-2221-E-006-187.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

energy harvesting, force sensors, PDMS, pressure, triboelectric nanogenerators

Received: February 25, 2020
Revised: May 30, 2020
Published online:

[1] a) H. J. Yoon, H. Ryu, S. W. Kim, *Nano Energy* **2018**, *51*, 270; b) W. Kim, D. Bhatia, S. Jeong, D. Choi, *Nano Energy* **2019**, *56*, 307; c) V. L. Trinh, C. K. Chung, *Appl. Energy* **2018**, *213*, 353;
[2] a) T. Quan, Y. Wu, Y. Yang, *Nano Res.* **2015**, *8*, 3272; b) X. Wen, Y. Su, Y. Yang, H. Zhang, Z. L. Wang, *Nano Energy* **2014**, *4*, 150; c) Y. Su, Y. Yang, X. Zhong, H. Zhang, Z. Wu, Y. Jiang, Z. L. Wang, *ACS Appl. Mater. Interfaces* **2014**, *6*, 553; d) X. Wang, Y. Yang, *Nano Energy* **2017**, *32*, 36.
[3] a) J.-H. Lee, J. Kim, T. Y. Kim, M. S. Al Hossain, S.-W. Kim, J. H. Kim, *J. Mater. Chem. A* **2016**, *4*, 7983; b) S. H. Wang, Z. L. Wang, Y. Yang, *Adv. Mater.* **2016**, *28*, 2881; c) R. Cao, J. N. Wang, Y. Xing, W. X. Song, N. W. Li, S. Y. Zhao, C. Zhang, C. J. Li, *Adv. Mater. Technol.* **2018**, *3*, 1700371; d) Z. L. Wang, *Mater. Today* **2017**, *20*, 74;
[4] a) K. Zhang, S. Wang, Y. Yang, *Adv. Energy Mater.* **2017**, *7*, 1601852; b) X. Wang, Z. L. Wang, Y. Yang, *Nano Energy* **2016**, *26*, 164; c) Y. Yang, Z. L. Wang, *Nano Energy* **2015**, *14*, 245.
[5] a) S. Wang, L. Lin, Z. L. Wang, *N Energy* **2015**, *11*, 436; b) A. Chandrasekhar, G. Khandelwal, N. R. Alluri, V. Vivekananthan, S. J. Kim, *ACS Sustainable Chem. Eng.* **2018**, *6*, 6110; c) K. Zhao, Z. L. Wang, Y. Yang, *ACS Nano* **2016**, *10*, 9044; d) Y. Yang, H. Zhang, Z. L. Wang, *Adv. Funct. Mater.* **2014**, *24*, 3745; e) Y. Wu, X. Wang, Y. Yang, Z. L. Wang, *Nano Energy* **2015**, *11*, 162.
[6] Y. Yang, H. Zhang, Z. H. Lin, Y. S. Zhou, Q. Jing, Y. Su, J. Yang, J. Chen, C. Hu, Z. L. Wang, *ACS Nano* **2013**, *7*, 9213;
[7] P. Bai, G. Zhu, Q. Jing, J. Yang, J. Chen, Y. Su, J. Ma, G. Zhang, Z. L. Wang, *Adv. Funct. Mater.* **2014**, *24*, 5807;
[8] X. Wang, H. Zhang, L. Dong, X. Han, W. Du, J. Zhai, Z. L. Wang, *Adv. Mater.* **2016**, *28*, 2896;
[9] X. Pu, M. Liu, X. Chen, J. Sun, C. Du, Y. Zhang, J. Zhai, W. Hu, Z. L. Wang, *Sci. Adv.* **2017**, *3*, e1700015;

 © 2020 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

[10] I. W. Tcho, W. G. Kim, S. B. Jeon, S. J. Park, B. J. Lee, H. K. Bae, D. Kim, Y. K. Choi, *Nano Energy* **2017**, *42*, 34;
[11] S. Karmakar, P. Kumbhakar, K. Maity, D. Mandal, P. Kumbhakar, *Nano Energy* **2019**, *63*, 103831.
[12] a) M. Ma, Z. Kang, Q. Liao, Q. Zhang, F. Gao, X. Zhao, Z. Zhang, Y. Zhang, *Nano Res.* **2018**, *11*, 2951; b) A. Yu, Y. Zhu, W. Wang, J. Zhai, *Adv. Funct. Mater.* **2019**, *29*, 1900098.
[13] F. R. Fan, L. Lin, G. Zhu, W. Wu, R. Zhang, Z. L. Wang, *Nano Lett.* **2012**, *12*, 3109.
[14] W. Seung, M. K. Gupta, K. Y. Lee, K. S. Shin, J. H. Lee, T. Y. Kim, S. Kim, J. Lin, J. H. Kim, S. W. Kim, *ACS Nano* **2015**, *9*, 3501.
[15] W. G. Kim, I. W. Tcho, D. Kim, S. B. Jeon, S. J. Park, M. L. Seol, Y. K. Choi, *Nano Energy* **2016**, *27*, 306.
[16] G. G. Cheng, S. Y. Jiang, K. Li, Z. Q. Zhang, Y. Wang, N. Y. Yuan, J. N. Ding, W. Zhang, *Appl. Surf. Sci.* **2017**, *412*, 350.
[17] V. L. Trinh, C. K. Chung, *Small* **2017**, *13*, 1700373.
[18] a) W. Yang, X. Wang, H. Li, J. Wu, Y. Hu, *Nano Energy* **2018**, *51*, 241; b) W. Yang, X. Wang, H. Li, J. Wu, Y. Hu, Z. Li, H. Liu, *Nano Energy* **2019**, *57*, 41; c) M. L. Seol, S. H. Lee, J. W. Han, D. Kim, G. H. Cho, Y. K. Choi, *Nano Energy* **2015**, *17*, 63;
[19] F. Wortmann, K. Flüchter, *Bus. Inform. Syst. Eng.* **2015**, *57*, 221.
[20] C. K Chung, K. H. Ke, *Appl. Surf. Sci.* **2020**, *508*, 145310.
[21] B. A. Grzybowski, M. Fialkowski, J. A. Wiles, *J. Phys. Chem. B* **2005**, *109*, 20511.
[22] H. T. Baytekin, A. Z. Patashinski, M. Branicki, B. Baytekin, S. Soh, B. A. Grzybowski, *Science* **2011**, *333*, 308.

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